F79MA: Statistical Models A

Exam Solutions 2020-21

1. a) (similar to class exercises)

Given that observations are i.i.d. conditionally on θ, we have the likelihood

$$L(\theta;\underline{\mathrm{X}}) = \prod_{i=1}^{n} f(X_i;\theta) = \prod_{i=1}^{n} k\theta^{-1}X_i^{k-1}\exp\left(-X_i^k/\theta\right), \quad \underline{\mathrm{X}} \in {\mathbb{R}_+}^n,$$
$$= k^n\left(\prod_{i=1}^{n} X_i^{k-1}\right)\theta^{-n}\exp\left(-\sum_{i=1}^{n} X_i^k/\theta\right), \quad \underline{\mathrm{X}} \in {\mathbb{R}_+}^n$$

[2]

b) (similar to class exercises)

The likelihood can be factorised as follows

$$L(\theta;\underline{\mathrm{X}}) = h_1(\underline{\mathrm{X}})h_2(\theta, \sum_{i=1}^{n} X_i^k)$$

with

$$h_1(\underline{\mathrm{X}}) = k^n\left(\prod_{i=1}^{n} X_i^{k-1}\right)$$

and

$$h_2(\theta, T(\underline{\mathrm{X}})) = \theta^{-n}\exp\left(-\frac{1}{\theta}T(\underline{\mathrm{X}})\right)$$

with $T(\underline{\mathrm{X}}) = \sum_{i=1}^{n} X_i^k$. From the factorisation theorem, $T(\underline{\mathrm{X}})$ is sufficient for θ.

[2]

c) (similar to class exercise)

By taking the logarithm and differentiating w.r.t. θ we find that the score function

$$U(\theta) = -\frac{n}{\theta^2}\left(\theta - T(\underline{\mathrm{X}})/n\right)$$

It follows from the CRLB Corollary that $\hat{\theta} = T(\underline{\mathrm{X}})/n$ is the MLE.

[4]

d) (similar to class exercise)

It follows from the CRLB Corollary that $\hat{\theta} = T(\underline{\mathrm{X}})/n$ is unbiased. [3]

e) (similar to class exercise)

From the CRLB Corollary, $\hat{\theta}$ attains the CRLB for unbiased estimators of θ. So Var $\left(\hat{\theta}\right) = \theta^2/n$.

[3]

2. a) (similar to class exercises)

Given that observations are i.i.d. conditionally on λ, we have the likelihood

$$\begin{aligned} L(\lambda; \underline{\mathrm{X}}) &= \prod_{i=1}^{n} f(X_i; \lambda) = \prod_{i=1}^{n} \lambda \exp\left(-\lambda X_i\right), \quad \underline{\mathrm{X}} \in {\mathbb{R}_+}^n, \\ &= \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} X_i\right), \quad \underline{\mathrm{X}} \in {\mathbb{R}_+}^n \end{aligned}$$

[2]

b) (from lecture notes)

$$\begin{aligned} I(\lambda) &= -\mathrm{E}\left(\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2} \log L(\lambda; \underline{\mathrm{X}})\right), \\ &= \frac{n}{\lambda^2}. \end{aligned}$$

[2]

c) (from lecture notes)

$$\begin{aligned} \pi_J(\lambda) &\propto \sqrt{I(\lambda)}, \\ &\propto \lambda^{-1}, \qquad \lambda > 0. \end{aligned}$$

[1]

d) (class exercise)

From Bayes' theorem $p(\lambda|X_1, \ldots, X_n) \propto \lambda^{n-1} \exp\{(-\lambda \sum_{k=1}^{n} X_i)\}$ [2]

and hence $\lambda|x_1, \ldots, x_n \sim$ Gamma$(n, \sum_{k=1}^{n} X_i)$. [1]

The resulting posterior mean and variance are

$$\mathrm{E}(\lambda|\underline{X}) = \frac{n}{\sum_{k=1}^{n} X_i},$$

and

$$\text{Var}(\lambda|\underline{X}) = \frac{n}{\left(\sum_{k=1}^{n} X_i\right)^2}\,.$$

[1]

e) (unseen)

In order to calculate the bias of $\hat{\theta}(\underline{\mathrm{X}}) = \mathrm{E}(\lambda|\underline{X}) = \frac{n}{\sum_{k=1}^{n} X_i}$ we consider that the exponential sample $\underline{\mathrm{X}}$ is generated by using a fixed (unknown) value $\lambda_0 > 0$ and calculate

$$B(\lambda_0) = \mathrm{E}_{\underline{\mathbf{X}}|\lambda_0}\left(\hat{\theta}(\underline{\mathrm{X}})\right) - \lambda_0 = \mathrm{E}_{\underline{\mathbf{X}}|\lambda_0}\left(\frac{n}{\sum_{k=1}^{n} X_i}\right) - \lambda_0\,.$$

To calculate the above expectation, we use the linearity property of the expectation operator together with two properties of gamma random variables. First, that $X_i \sim \text{Gamma}(1, \lambda_0)$ for all $i = 1, \ldots, n$ and any $\lambda_0 > 0$, conditionally independent given λ_0, implies that $Y = \sum_{k=1}^{n} X_i \sim \text{Gamma}(n, \lambda_0)$.

Second, that for $Y \sim \text{Gamma}(n, \lambda_0)$, the expectation $\mathrm{E}(Y^{-1}) = \frac{\lambda_0}{n-1}$ for any $\lambda_0 > 0$.

Therefore,

$$B(\lambda) = \frac{n}{n-1}\lambda - \lambda = \frac{\lambda}{n-1}\,.$$

[5]

(solutions missing details or with minor errors receive half marks [2.5]; solutions with significant errors or that are incomplete receive not marks [0]).

f) (similar to class exercises)

The CRLB for $\hat{\theta}(\underline{\mathrm{X}})$ is given by

$$\frac{(1 + \frac{\mathrm{d}}{\mathrm{d}\lambda}B(\lambda))^2}{I(\lambda)} = \frac{\lambda^2}{n}\frac{n^2}{(n-1)^2} = \lambda^2\frac{n}{(n-1)^2}$$

where we recall that $I(\lambda) = n/\lambda^2$.

[2]

3. a) (lectures)

Given a class of tests for testing $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \notin \Theta_0$, a test in this class is said to be the uniformly most powerful test if, for all $\theta \notin \Theta_0$, it achieves a power that is greater or equal than that of any other test in the class. [2 Marks]

b) (similar to class example)

To show that the monotone likelihood ratio property is satisfied we construct the

likelihood ratio function

$$\begin{aligned}\frac{L(\mu_0)}{L(\mu_1)} &= \exp\left\{-\frac{1}{2\sigma^2}\left(\sum x_i^2 - 2\mu_0\sum x_i + n\mu_0^2\right) + \frac{1}{2\sigma^2}\left(\sum x_i^2 - 2\mu_1\sum x_i + n\mu_1^2\right)\right\} \\ &= \exp\left\{-\frac{1}{\sigma^2}(\mu_1-\mu_0)\sum x_i + \frac{n}{2\sigma^2}(\mu_1^2-\mu_0^2)\right\}.\end{aligned}$$

[2 Marks]

The likelihood ratio is a monotone decreasing function of the statistic $\sum x_i$ (since $\mu_1 > \mu_0$), and hence

$$R = \left\{\underline{\mathrm{x}} : \sum x_i \geq k\right\}.$$

[1 Mark]

Alternatively, it is also possible to express R as follows

$$R = \{\underline{\mathrm{x}} : \bar{x} \geq k_1\}$$

or

$$R = \left\{\underline{\mathrm{x}} : \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \geq k_2\right\}.$$

c) (similar to class examples)

To determine the reject region for a test of size $\alpha = 0.05$ we use the sampling distribution of $\bar{X}$ at $\mu_0 = 150$.

More precisely, we use

$$\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

[2 Marks]

and obtain the value $k_2 = 1.6449$ for a test of size $\alpha = 0.05$, $\mu_0 = 150$, $\sigma = 100$, and $n = 5$. Similarly,

$$k_1 = 150 + 1.6445(100/\sqrt{5}) = 223.54; \quad k = nk_1 = 1117.7.$$

[1 Marks]

d) (similar to class example)
The power of the test is

$$\Pr(\bar{X} \geq 223.54|H_1).$$

[1 Mark]

For $\mu = 300$ we have

$$\begin{aligned}1-\beta &= \Pr\left(\sqrt{5}\frac{\bar{X}-300}{100} \geq \sqrt{5}\frac{223.54-300}{100}\right) \\ &= \Pr(Z \geq -1.71) = 0.9564\,.\end{aligned}$$

[1 Marks]

e) (from lecture notes and standard example)
The p-value is defined as the minimum level of test at which H_0 would be rejected, or alternatively as the probability of observing data as extreme as the given data when H_0 is true. [1 Marks]
The p-value for $\bar{x} = 160$ is given by:

$$\Pr\left(Z > \sqrt{5}\frac{160-150}{100}\right) = \Pr(Z > 0.2236) = 0.4115 > 0.05.$$

We do not have sufficient evidence to reject the null hypothesis. [2 Mark]

4. (from lecture notes)

a) The posterior density of λ given $\underline{x}$ is proportional to

$$\begin{aligned} f(\lambda|\underline{x}) &\propto \prod_{i=1}^{n} \frac{\lambda^{x_i}}{x_i!} e^{-\lambda} \frac{\beta^{\alpha}}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda} \\ &\propto \lambda^{\alpha+\sum_i x_i - 1} e^{-(\beta+n)\lambda}, \end{aligned}$$

which is proportional to the pdf of a gamma$(\alpha + \sum_i x_i, \beta + n)$. [2 Marks]

$E[\lambda|\underline{x}]$ is the mean of the posterior distribution, i.e. Bayes estimate with respect to quadratic loss function. [1 Mark]

$$E[\lambda|\underline{x}] = \left(\alpha + \sum_{i=1}^{n} x_i\right)/(\beta + n).$$

[2 Marks]

b) (from lecture notes)

$$E[\lambda|\underline{x}] = Z\left\{\sum_{i=1}^{n} x_i/n\right\} + (1-Z)\alpha/\beta$$

[2 Marks]

where $Z = n/(\beta + n)$. [1 Mark]

c) (from lecture note)

Credibility premium is a linear function of the observed values. Z depends on the amount of data available for the risk, n, and collateral information, through β. As n increases the sampling error of $\bar{X}$ as an estimator for λ decreases. Similarly, β reflects the variance of the prior distribution for λ. Thus Z reflects the relative reliability of the two estimates of λ.

[2 Marks]

If $n = 0$, the credibility factor, i.e. $Z = 0$. In this case, the only information we have is the prior distribution and the best estimate of λ is the mean of the prior

distribution, which is α/β. On the other hand, if we can only estimate λ using the data from the risk itself, the estimate for λ would be $\sum_{i=1}^{n} x_i/n$ which is the maximum likelihood estimate of λ.

[2 Marks]

d) (from lecture notes and standard example)

$m(\theta) = E[X_j|\theta]$, so

$$E[m(\theta)] = (88.6 + 71 + 65.2 + 67.8)/4 = 73.15.$$

[1 Mark]

$$E[s^2(\theta)] = (89.8 + 280 + 722.7 + 474.2)/4 = 391.68.$$

[1 Mark]

The between risk variance is

$$\left((88.6 - 73.15)^2 + (71 - 73.15)^2 + (65.2 - 73.15)^2 + (67.8 - 73.15)^2\right)/3 = 111.72.$$

111.72 - 391.68/5 =33.38

[1 Mark]

The credibility factory, Z is

$$\begin{aligned} Z &= \frac{n}{n + E[s^2(\theta)]/V[m(\theta)]} \\ &= \frac{5}{5 + 391.68/33.38} = 0.2988. \end{aligned}$$

[1 Mark]

The credibility premium for the coming year for risk number 1 is

$$0.2988 \times 88.6 + (1 - 0.2988) \times 73.15 = 77.77.$$

[1 Mark]